# Instructor key: Paper 010

**AuthorityEffect:** None. Use after the quiz. A plausible label alone earns at most one of each question's three points.

## Levels 1–2

1. i j (−i) (−j)=k(−i)(−j)=(−j)(−j)=−1. Reordering to i(−i)j(−j) gives 1.
2. Associativity supports group/path reasoning and is not implied by totality. S3 has $6^3$=216 triples, Q8 has $8^3$=512; validation costs O(|G|³).
3. `GroupTableError`: a list is not a declared element and may be unhashable. No verdict.
4. Reversal gives hol $^1$; rotation gives U01 $^1$ hol U01. Central self-inverse −1 stays unchanged under either operation.
5. Accept if it is an actual plaquette, its listed-order product is `r`, and it is nonidentity. The old harness accepted only one convention.
6. Adjacent g(v) $^1$g(v) cancel, leaving endpoint factors. Identity status and conjugacy class persist for a closed path; the exact element may change.

## Levels 3–4

7. Set g(root)=e and g(v)=g(u)Uuv along the tree. Flatness permits plaquette path substitutions, so path independence on the rectangle extends this relation to all edges. Each open dimension must be at least 2.
8. `FLAT_NONTRIVIAL`: nonidentity A cannot be conjugated to identity.
9. j i (−j)(−i)=−1, so the corner and connection are `NOT_FLAT`.
10. Skipping changes 20→0; flatness, gauge triviality and verdict persist.
11. Validation O(|G|³); lattice/tree/gauge work O(LM); a k-edge certificate path O(k).
12. Four identity links give identity. The named theorem has no axioms; the full historical report has 26 declarations, 25 axiom-free and one `propext`. The complete checker and empirical analysis are outside the proof.

## Levels 5–6

13. Reject: `gauge missing`. A matching label does not meet the witness obligation.
14. These spellings parse to one edge/vertex. Reject duplicates, even equal ones. Unknown/non-string links receive `LatticeError` before inverse lookup.
15. Accept an actual non-flat plaquette/product; reject a wrong product. Accept a full trivializing gauge; reject a missing vertex. Accept a flat torus with a matching closed nonidentity loop; reject an open or identity loop. A four-cycle can wind on a torus without bounding a plaquette.
16. 25 recorded responses; 15 retained solution records, 10 accepted and 5 rejected afresh. Thirteen records are parsed certificates, two are parser-

error wrappers. Ten missing payloads retain historical failures; their fresh checks are not run.

17. The messages establish convention-based rejection before product comparison, not a wrong product. Absent payloads cannot be reconstructed.
18. This shorter prompt produced an accepted response within its limit on this case. Several prompt changes, budgets, provider/sampling conditions and one reply per arm prevent a general savings or causal estimate.

## Levels 7–8

19. C049 changes cases (stealth 5→9), so it is not a matched repeat. C050 repeats ten cases (7→6); five outcomes change, two involving API errors. Tasks, small authored panels, leakage and absent statistical design preclude model ranking.
20. 7+5+4+10+5+5=36/60. This is the author's identifier classification, not a measured dependence of model replies on leaked answers.
21. Model counts do not exceed the refusal baseline. The second checker accepts vacuous/unsupported outputs in the cited controls; agreement alone supplies no general independent assurance.
22. C017 replay executes algebra/certificate checks. Table 3 recount reads old flags; Table 4 recount reads annotations/prior checker outcomes. A fresh Lean build invokes the pin. A new trial obtains new provider responses.
23. Membership and byte identity are checked. Correctness, proof scope, design and rendering need separate evidence. Record actual failed/not-run toolchain status and retain earlier successful receipts as historical.
24. Use held-out cases with opaque IDs and stripped notes; repair/check refusal rules and both checkers; match conditions/budgets; record providers, errors and repeated samples; predeclare comparisons and retain complete outputs. This proposes work rather than reporting its execution.

## Capstone rubric: eight points

One point for each audit item named in the quiz, supported by a calculation, output field or source. S3 has 18 commuting seam pairs and Q8 has 40: 58 flat of 100, comprising 2 `FLAT_TRIVIAL`, 56 `FLAT_NONTRIVIAL`, and 42 `NOT_FLAT`. Gauge skipping is 20→0. C017 counts stay 15/10/5 with ten missing payloads. Historical table recounts preserve Tables 3/4 and do not execute the formal checkers. The capstone runs neither Lean nor a provider. An 80-point practice score or receipt grants no further authority.